

# EMERGENT TRAFFIC CONGESTION SIMULATION

Complex Systems - NetLogo Project Report

|      |         |            |                        |
|------|---------|------------|------------------------|
| Tool | NetLogo | Model Type | Agent-Based Simulation |
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## Project Overview

This project presents an agent-based traffic simulation developed in NetLogo to study how traffic congestion emerges from local interactions among vehicles and a traffic signal. Cars travel on a two-lane road and independently adjust their speed according to the vehicle ahead and the current traffic-light state. Congestion is not created by a central controller; queues and slowdowns emerge when many agents follow the same simple rules.

## Objectives

- Model traffic flow using autonomous vehicle agents.
- Demonstrate emergence in a complex system.
- Observe congestion-related behaviour under changing traffic conditions.

## Complex-System Features

- **Agents:** Cars and a traffic light
- **Local rules:** Car-following, safe spacing, acceleration/deceleration, and signal response
- **Emergence:** Queues, stop-and-go movement, and system-level congestion

## Model Behaviour and Measurements

- **Vehicle behaviour:** Cars detect the nearest vehicle ahead in the same lane, maintain spacing, and smoothly accelerate or decelerate.
- **Traffic signal:** The signal cycles through green, yellow, and red; approaching cars continue, slow, or stop accordingly.
- **Measurements:** Average speed, stopped cars, traffic density, congestion level, queue length, and completed trips are monitored to evaluate system behaviour.

## Team Members and Contributions

| No | Member Name             | Student ID | Main Contribution                                          |
|----|-------------------------|------------|------------------------------------------------------------|
| 1  | H.K.S. Peiris           | FC212045   | Model concept, problem definition, and simulation setup    |
| 2  | J.M.D.M. Sulakshan      | FC212026   | NetLogo implementation and vehicle movement logic          |
| 3  | W.A.Y.L. Wickramasinghe | FC212004   | Traffic-light logic, testing, and result analysis          |
| 4  | H.E.M.U. Simal          | FC212025   | Documentation, presentation preparation, and video editing |